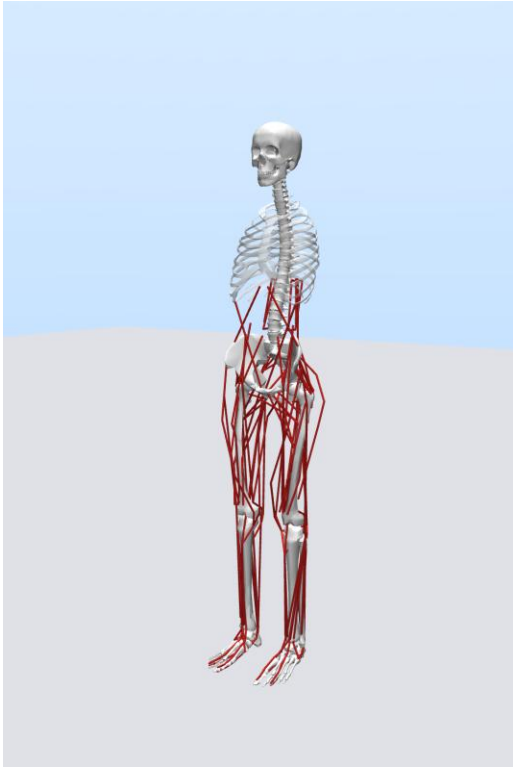
PHD DEFENSE

Movement and Control of Biomimetic Humanoid Robots

Ben P. Bolen

Mechanical Engineering

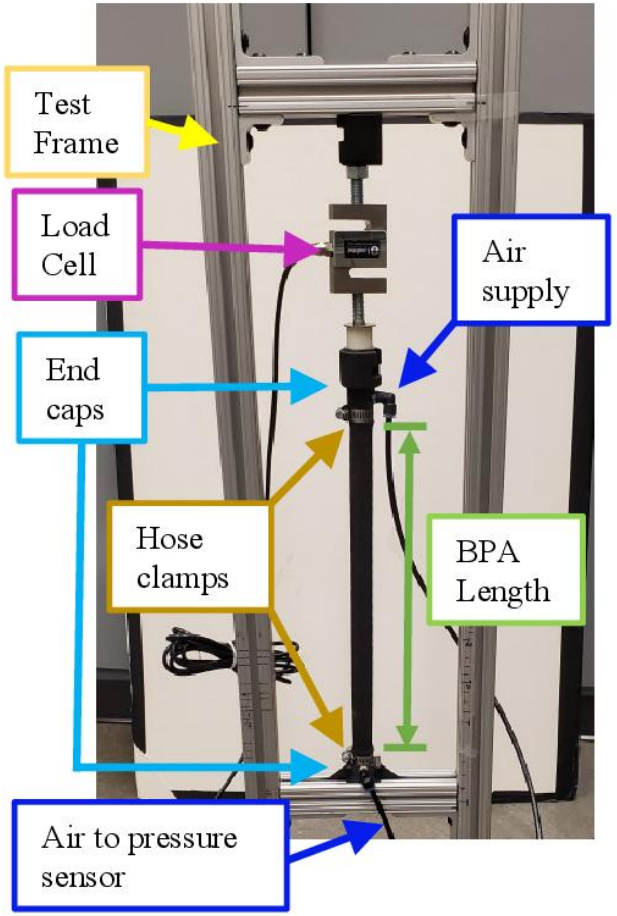
November 6, 2026



A research robot needs evidence at each model boundary

Unknown in the design	Evidence developed here
Force when actuator length changes	Isometric measurements and force model
Torque through real attachments	Correction identification and route tests
Which sensory findings apply	Searchable, source-linked SADb app
What a coupled circuit actually does	Matched tests and controlled interventions

Isometric tests isolate force across BPA design choices



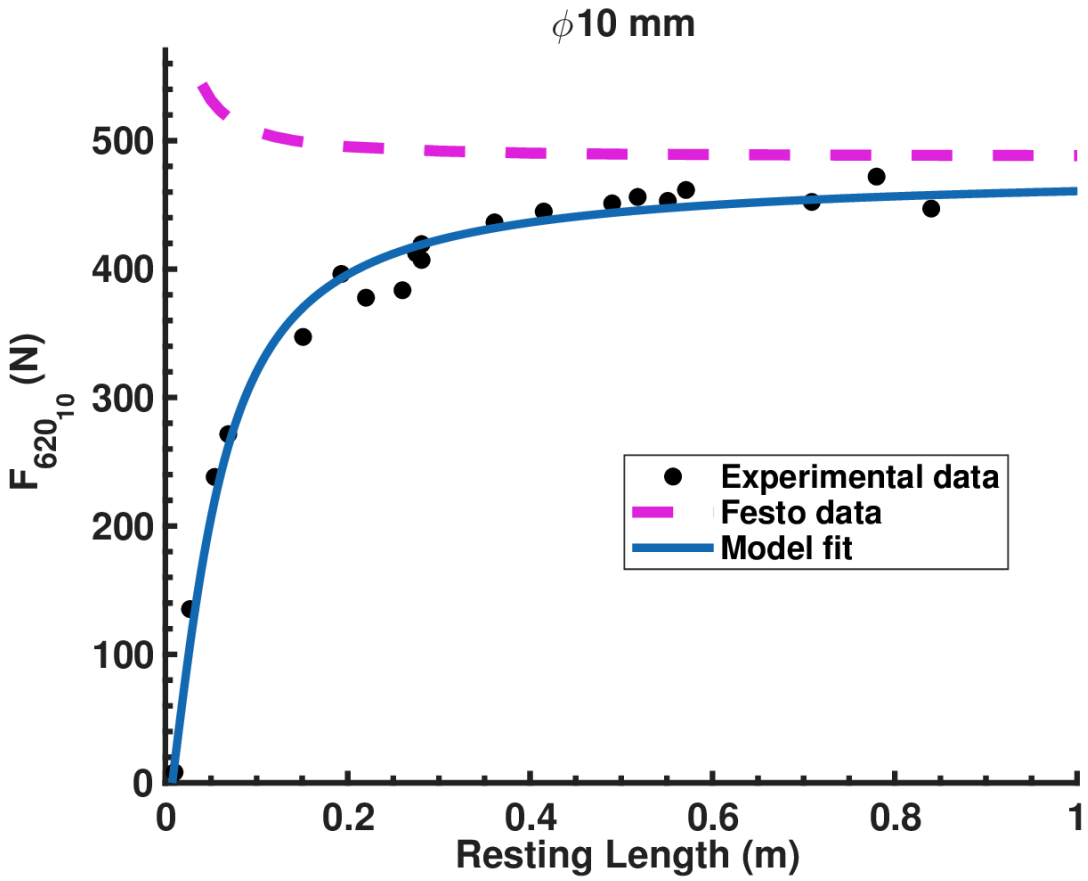
537 observations

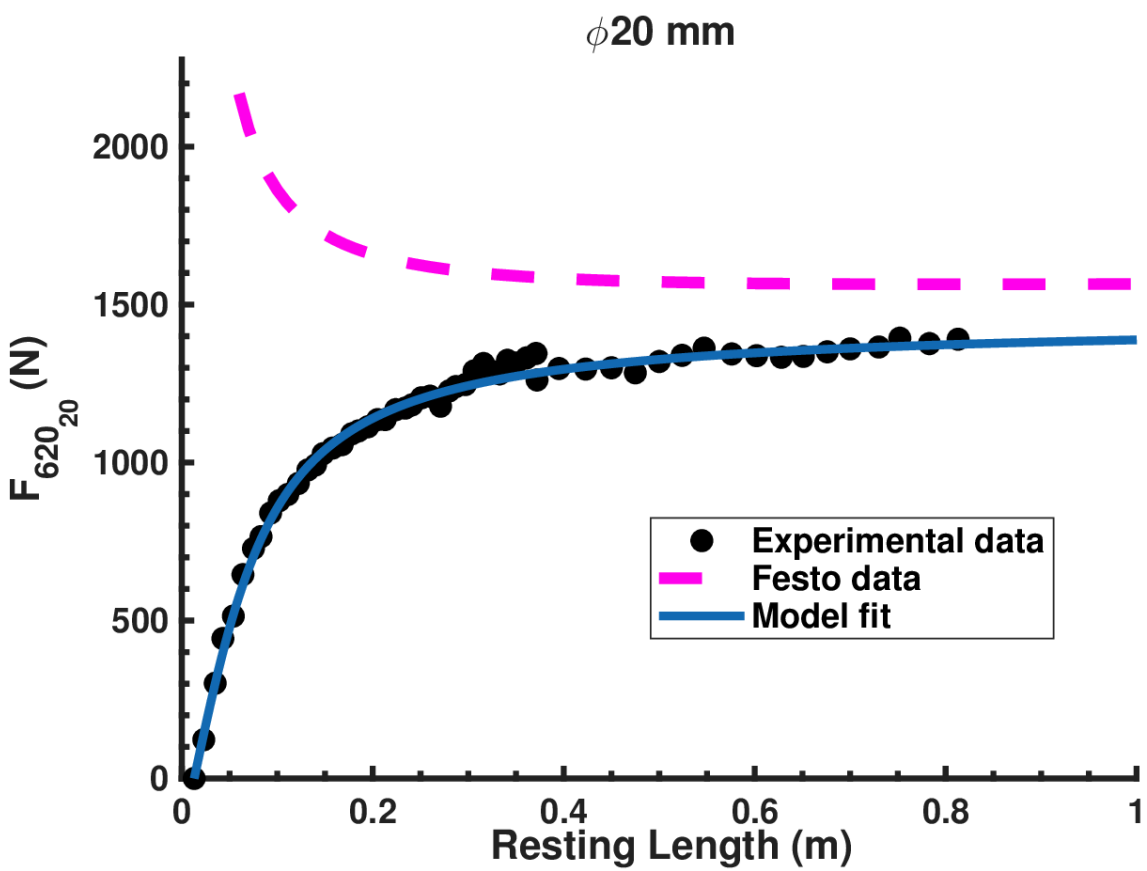
10 mm and 20 mm BPAs

Length, contraction, and pressure

Force measured at fixed length

Shorter BPAs deliver less maximum isometric force

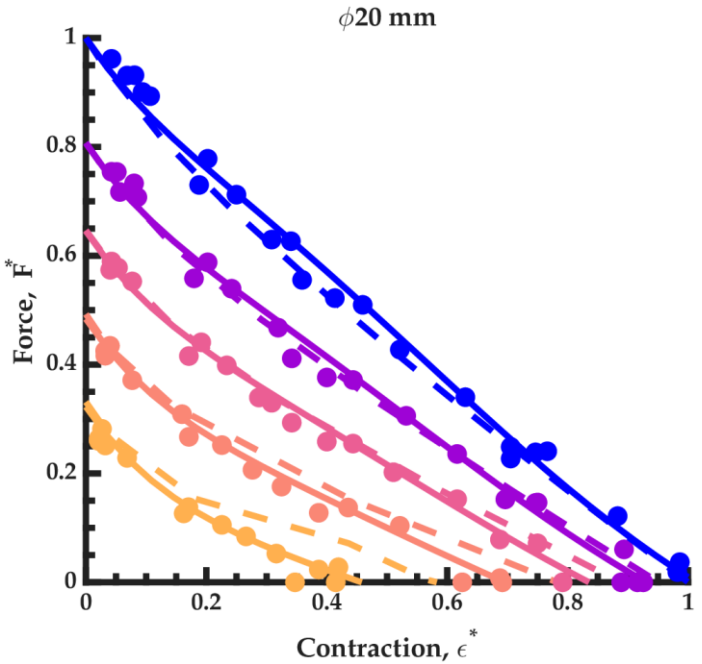




Resting length changes the force available to a robot design

Resting-length scaling completes the normalized force model

$$F(\varepsilon^*, P^*, l_0) = F^*(\varepsilon^*, P^*) \cdot F_{620}(l_0)$$

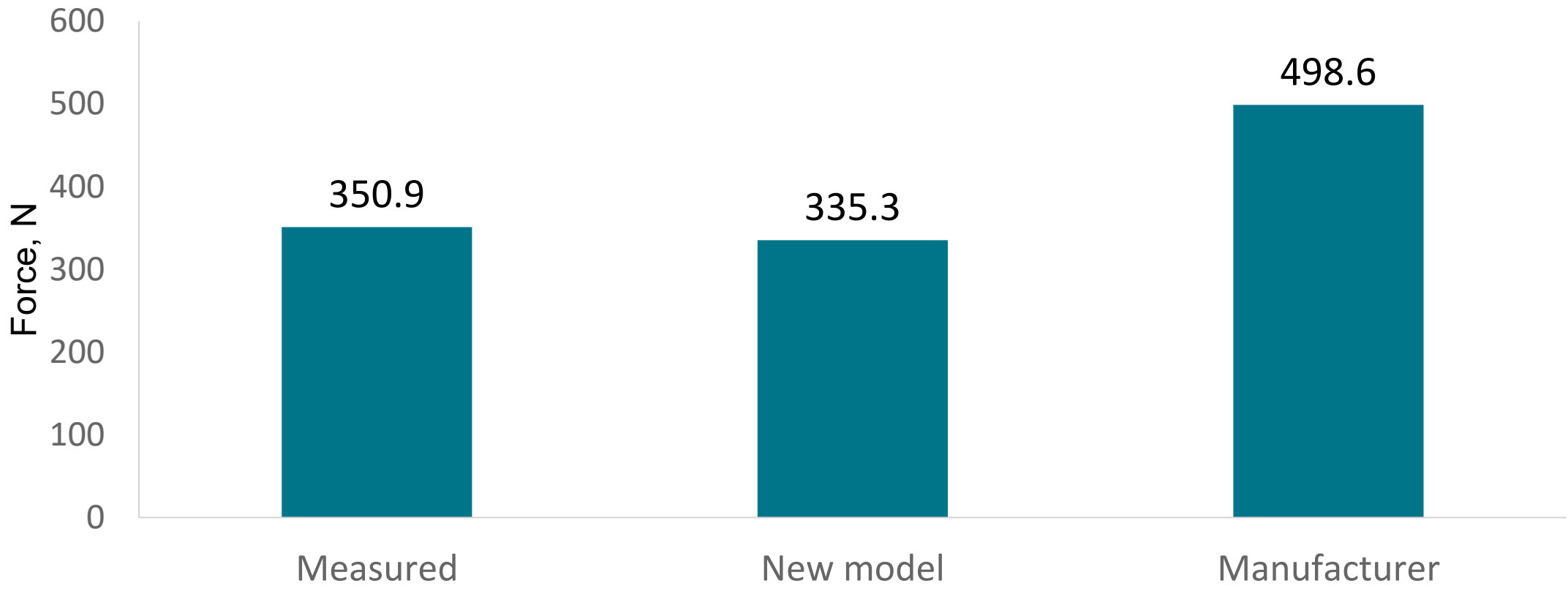


F* normalized force
P* pressure / 620 kPa
ε* normalized contraction
l_o resting length

Curves: 200–620 kPa
Solid model; circles measured
Dashed manufacturer

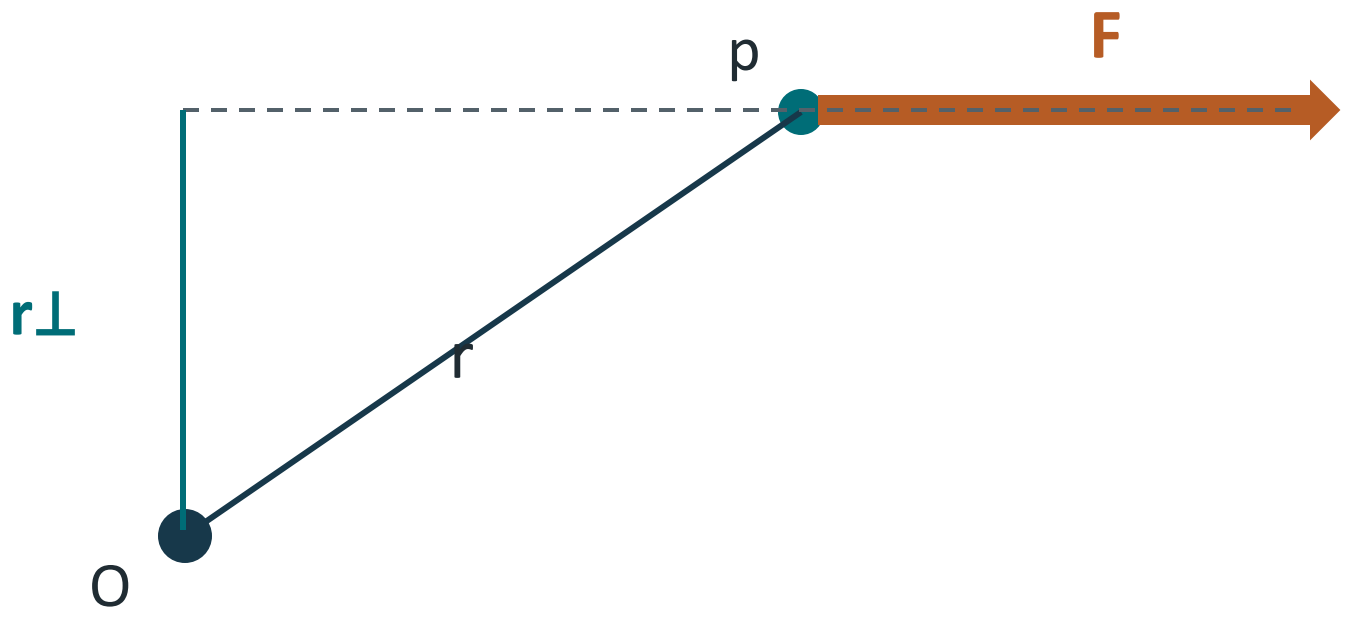
Diameter-specific coefficients remain fixed from the published study

At 112 mm, the new model reduces force error to 4.5%



10 mm BPA · 112 mm resting length · 620 kPa

Joint torque depends on force direction and moment arm



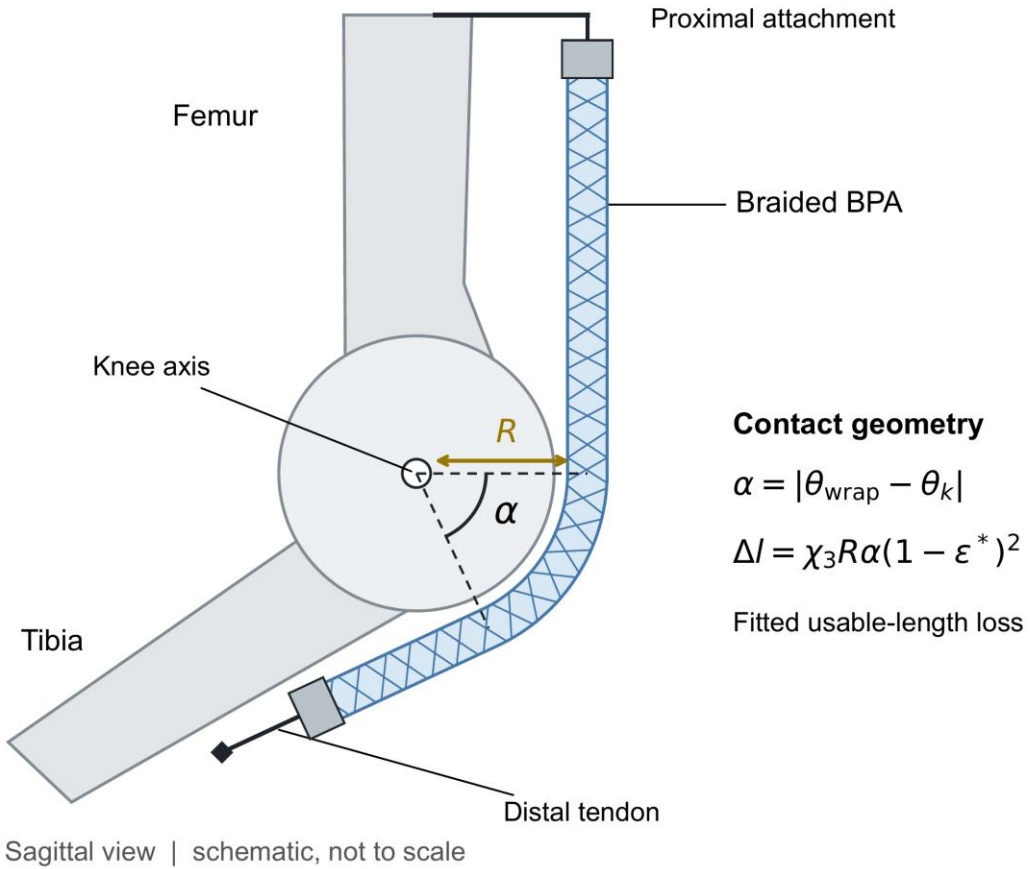
$$\tau_z = (r \times F) \cdot \hat{z}$$

$$|\tau_z| = F\, r_\perp$$

Routing also changes actuator length

A force model needs a transmission model to predict delivered torque

Length, compliance, and wrapping alter usable contraction



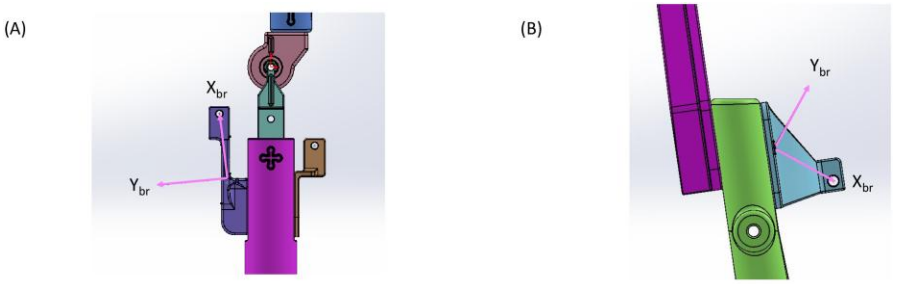
Sagittal view | schematic, not to scale

Contact geometry

$\alpha = |\theta_{wrap} - \theta_k|$

$\Delta l = \chi_3 R \alpha (1 - \varepsilon^*)^2$

Fitted usable-length loss



Constant offset

Load-dependent deflection

Contact-dependent length loss

The correction terms belong to the actuator–attachment system

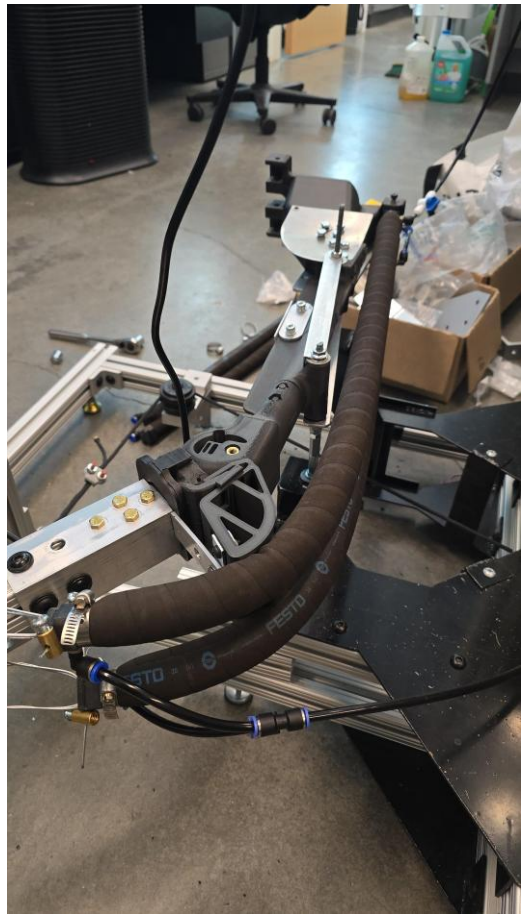
Corrections reduce error on held-out knee tests

Validation RMSE, N m	No corrections	Corrected
Flexor, 2 tests	7.588	1.490
Extensor, 3 tests	2.554	1.098

Same stiffness pair across both configurations

Tests reserved before evaluating the adopted fit

The corrected model guides a buildable two-BPA cam route



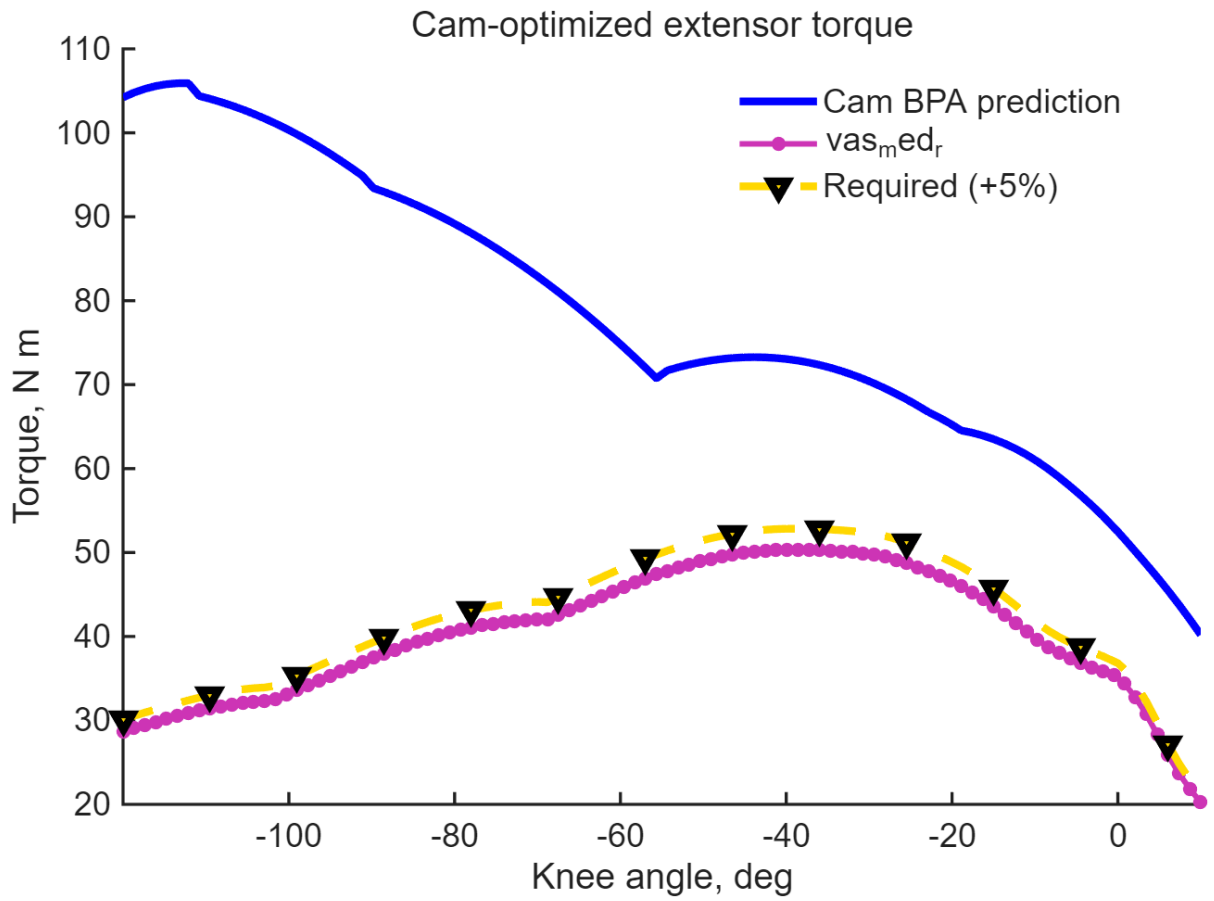
Two 20 mm BPAs

Cam-assisted tendon path

Adopted transmission corrections guide the design

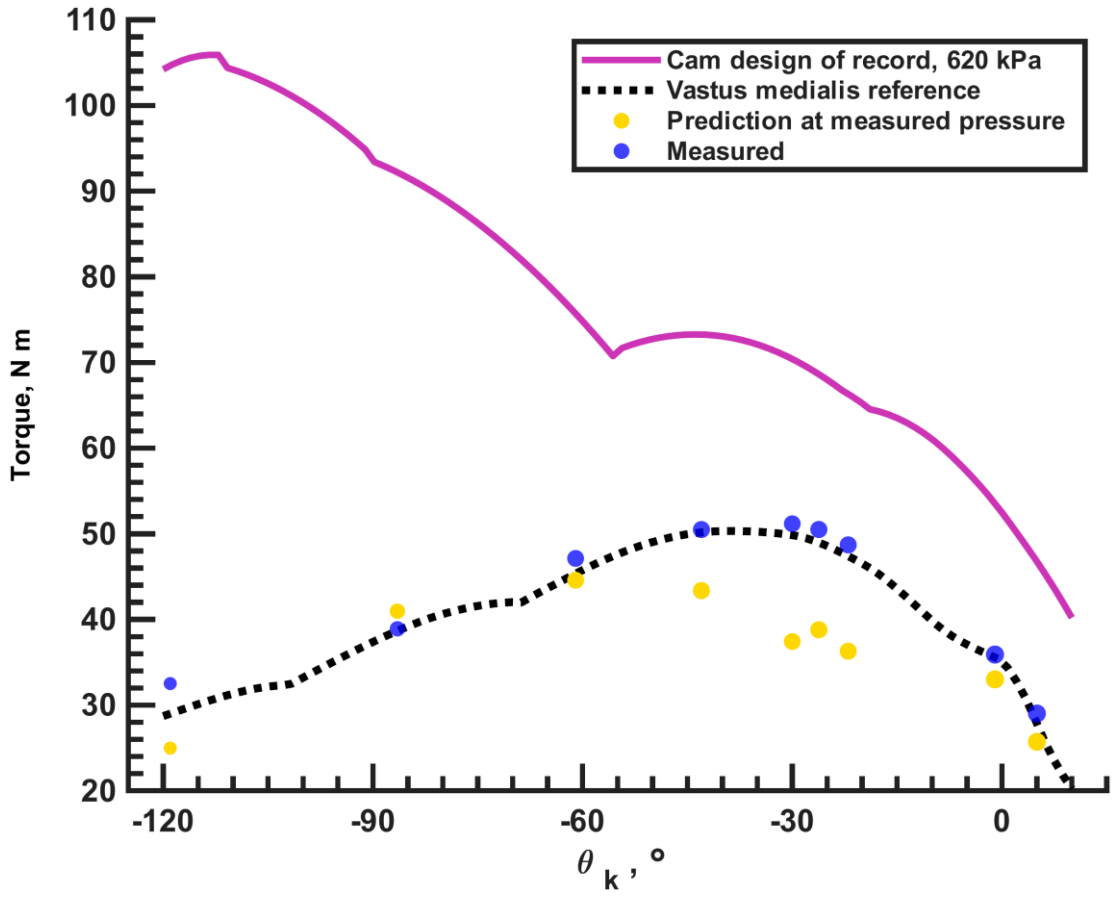
The route must satisfy force, excursion, and geometric constraints

The cam model predicts headroom above the required target



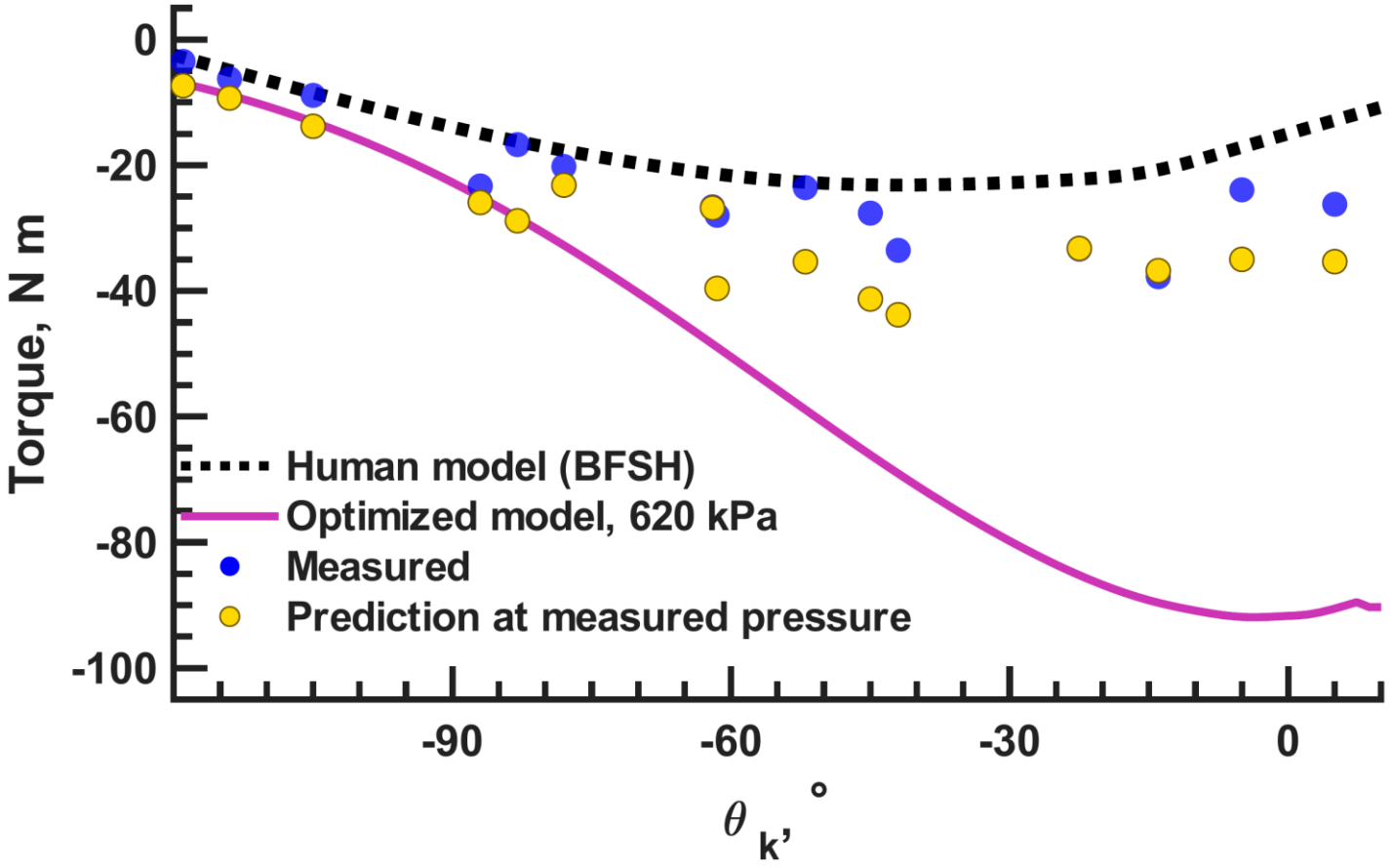
+30.83% above required envelope · 269–507 kPa predicted pressure

All nine extensor measurements meet the human target



9 of 9 target margins positive · +0.24 to +3.52 N m

All fifteen flexor measurements meet the human target



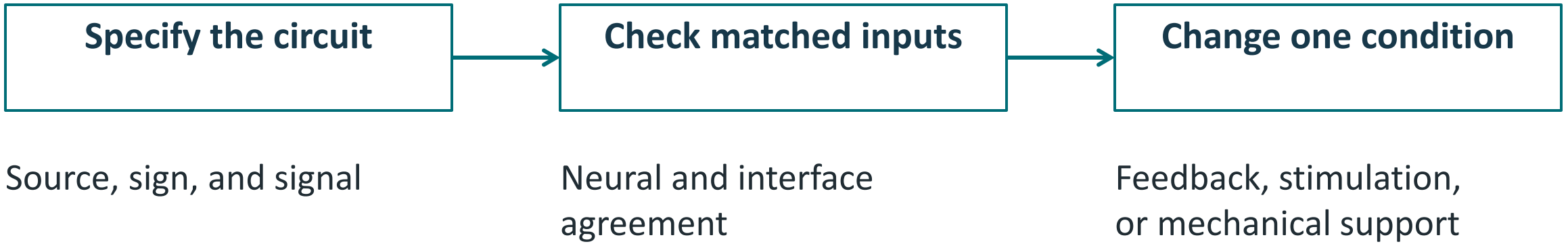
15 of 15 targets met · 226.5–638 kPa · negative torque denotes flexion

SADb makes sensory evidence traceable to its source context



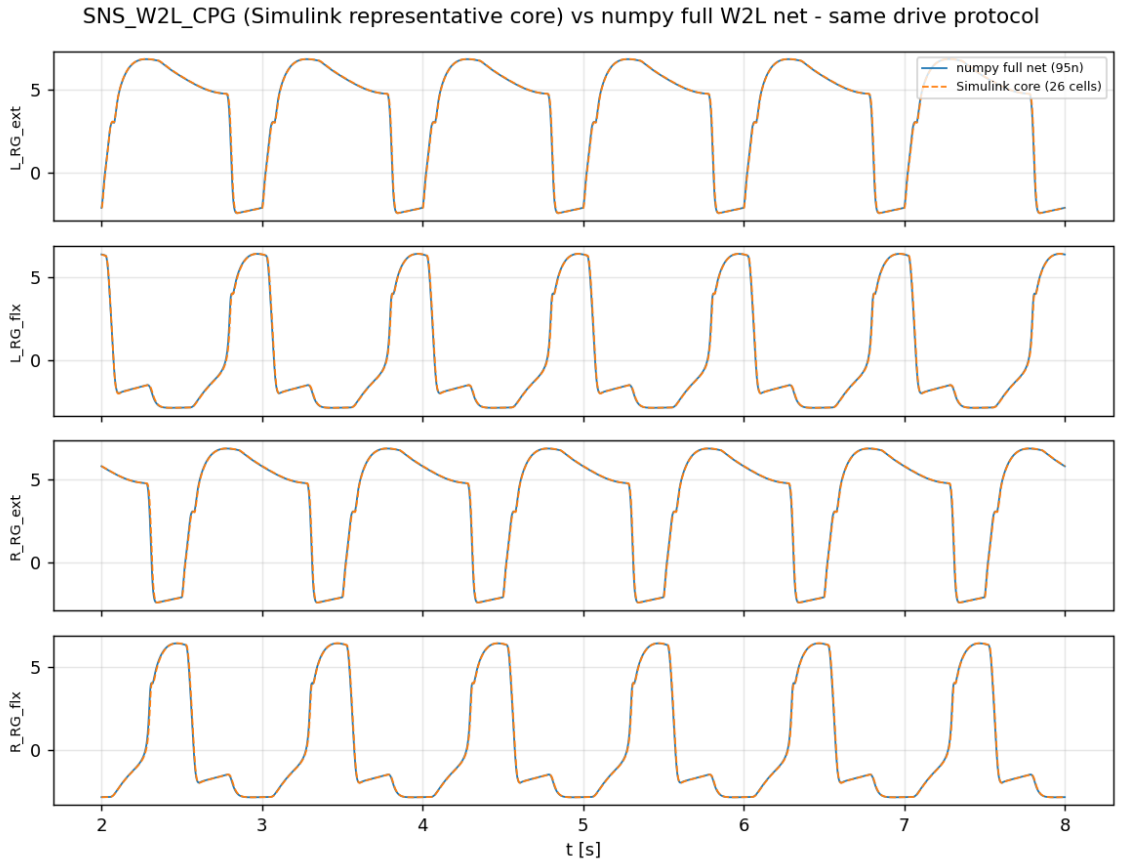
1,393 total records · search, pivot, and follow citation links

Controlled interventions turn a coupled model into an experiment



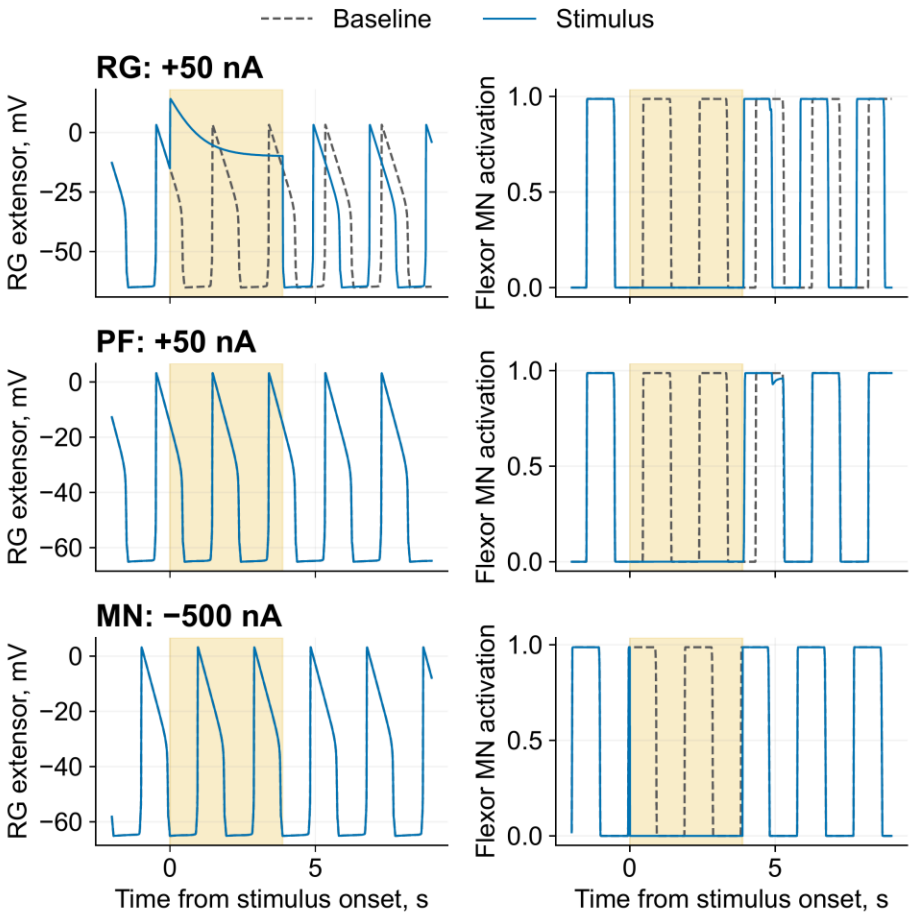
The demonstrations test implementation, sensory contribution, and mechanical response separately

Matched rhythm cores reproduce the same driven bursts



1.000 s period · 10 expected bursts · correlation ≥ 0.99996

Two deleted bursts can preserve or reset the rhythm

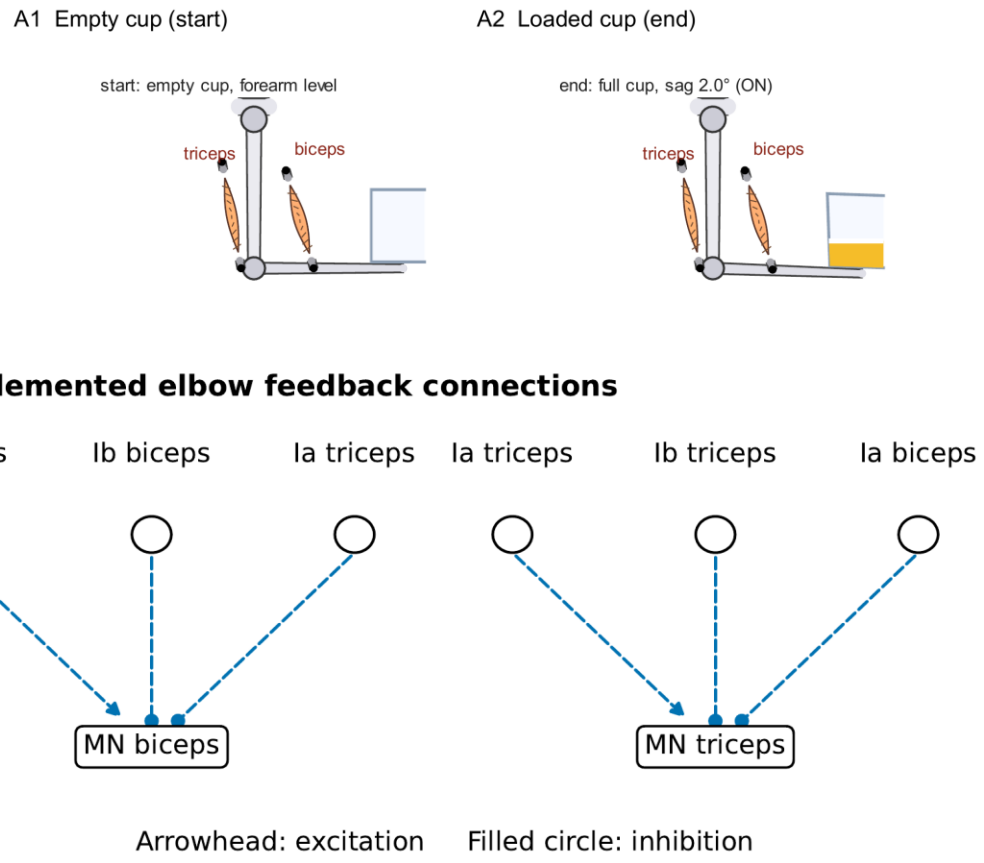


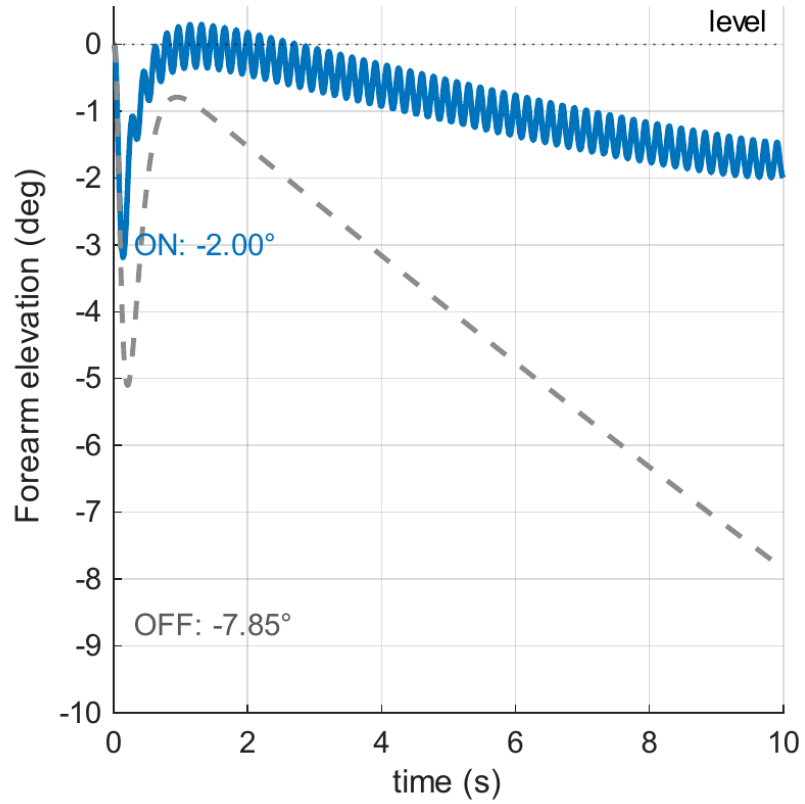
Stimulated layer	RG shift, cycles
RG	0.211
PF	0.000
MN	0.000

RG: rhythm generator
PF: pattern formation
MN: motor neuron

Reduced Deng-derived circuit · two baseline periods · suspended model

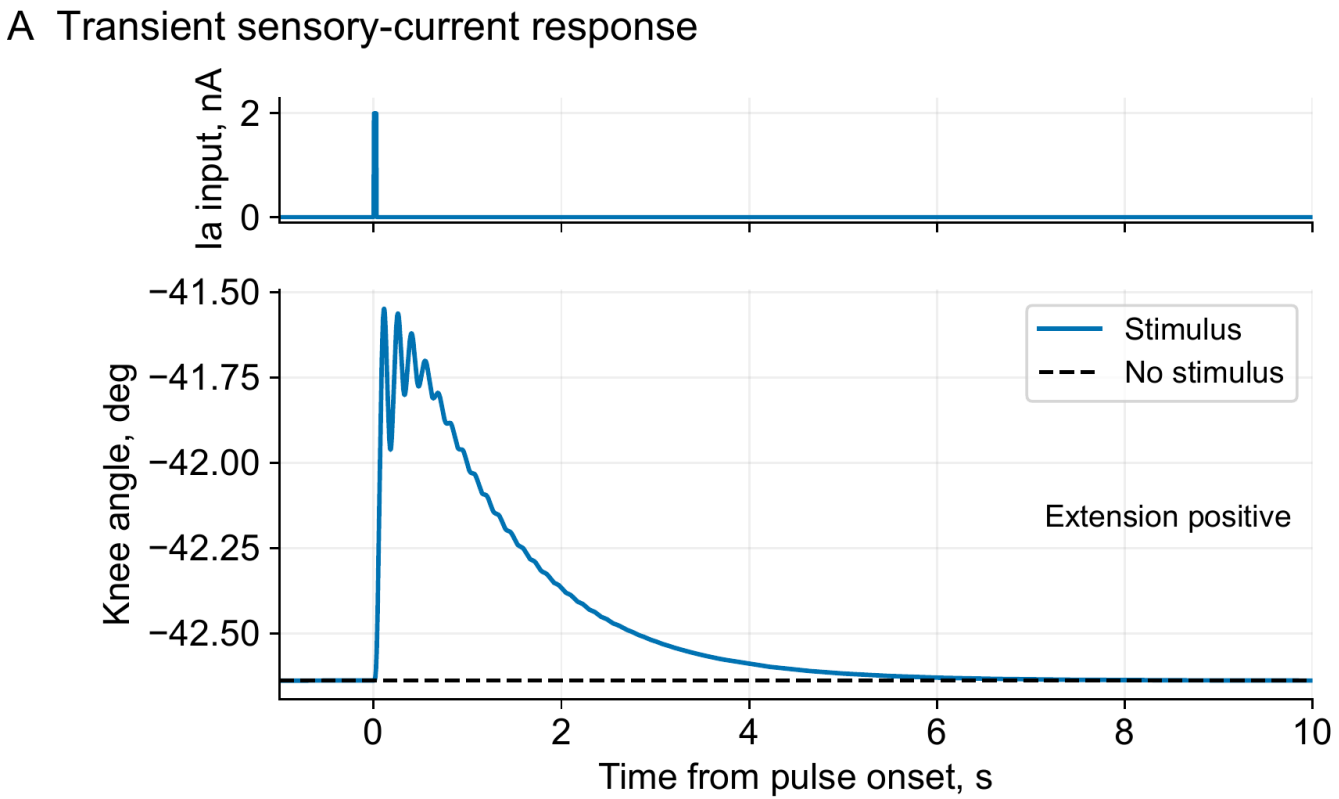
Sensory feedback reduces sag in the loaded-arm model

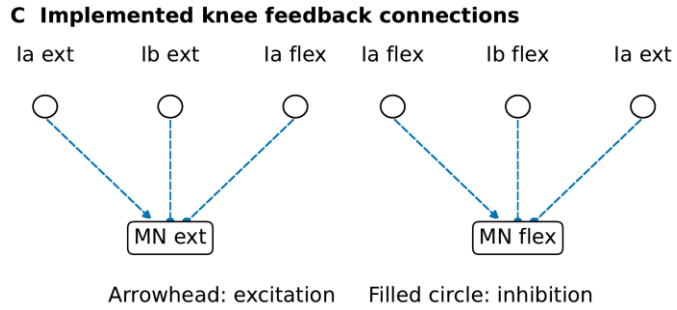




Sag: 2.00° with feedback · 7.85° without feedback

The knee model produces a transient sensory response





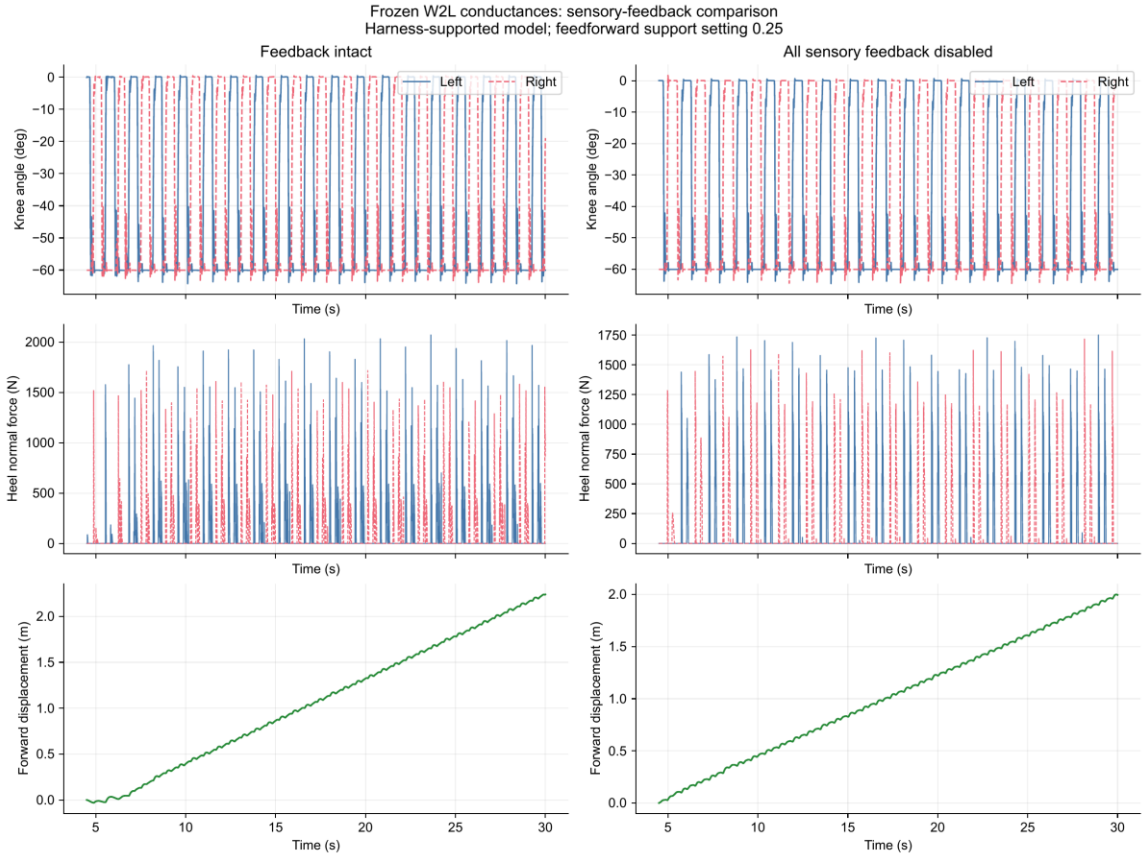
2 nA for 30 ms

1.09° at 111 ms

Recovery at 6.355 s

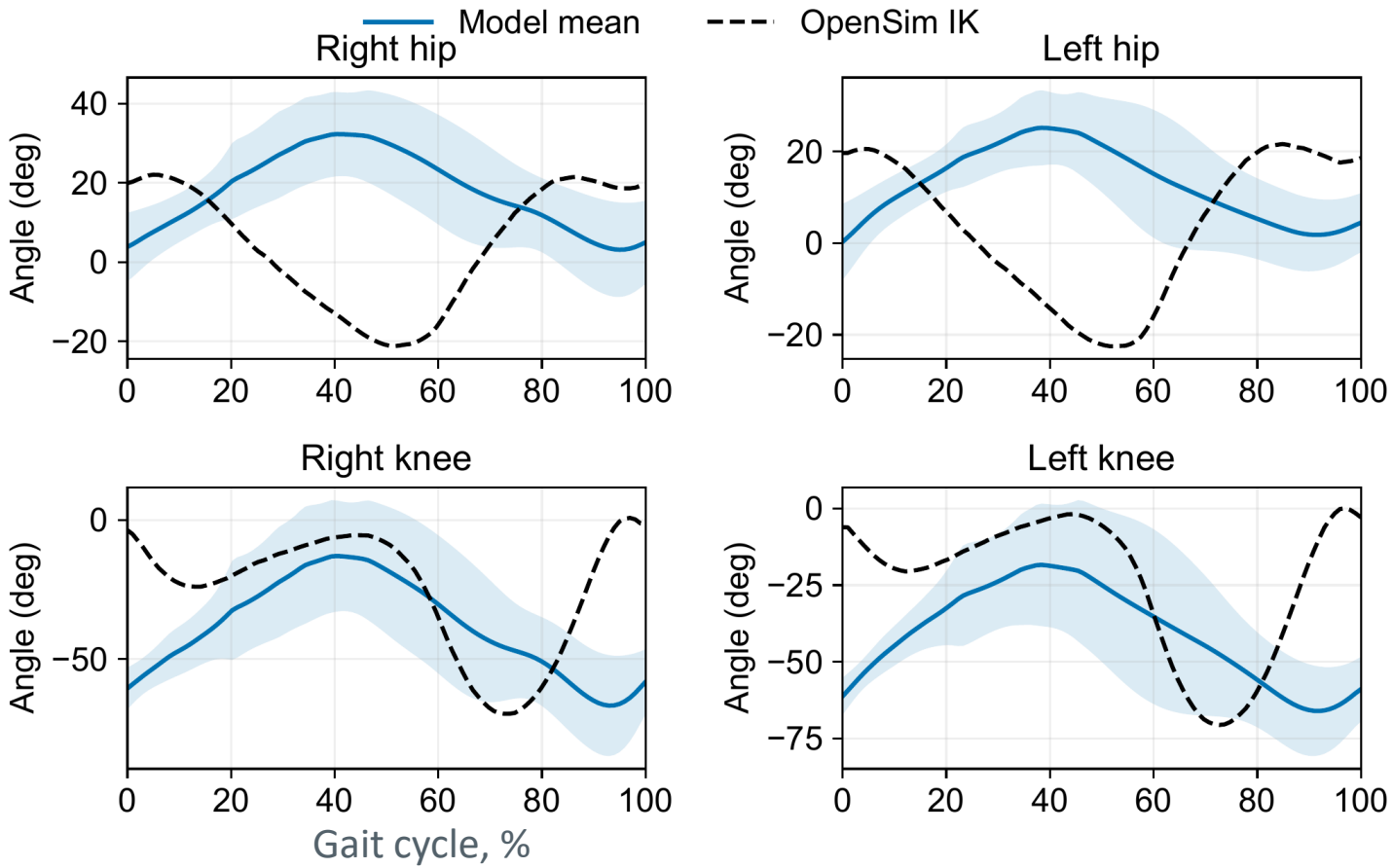
Six neural saturation widths changed in this configuration

Feedback coordinates foot loading in the W2L model



Median bilateral contact phase: 0.498 intact · 0.223 without feedback

Rhythmic Gait2392 cycles retain hip and knee errors



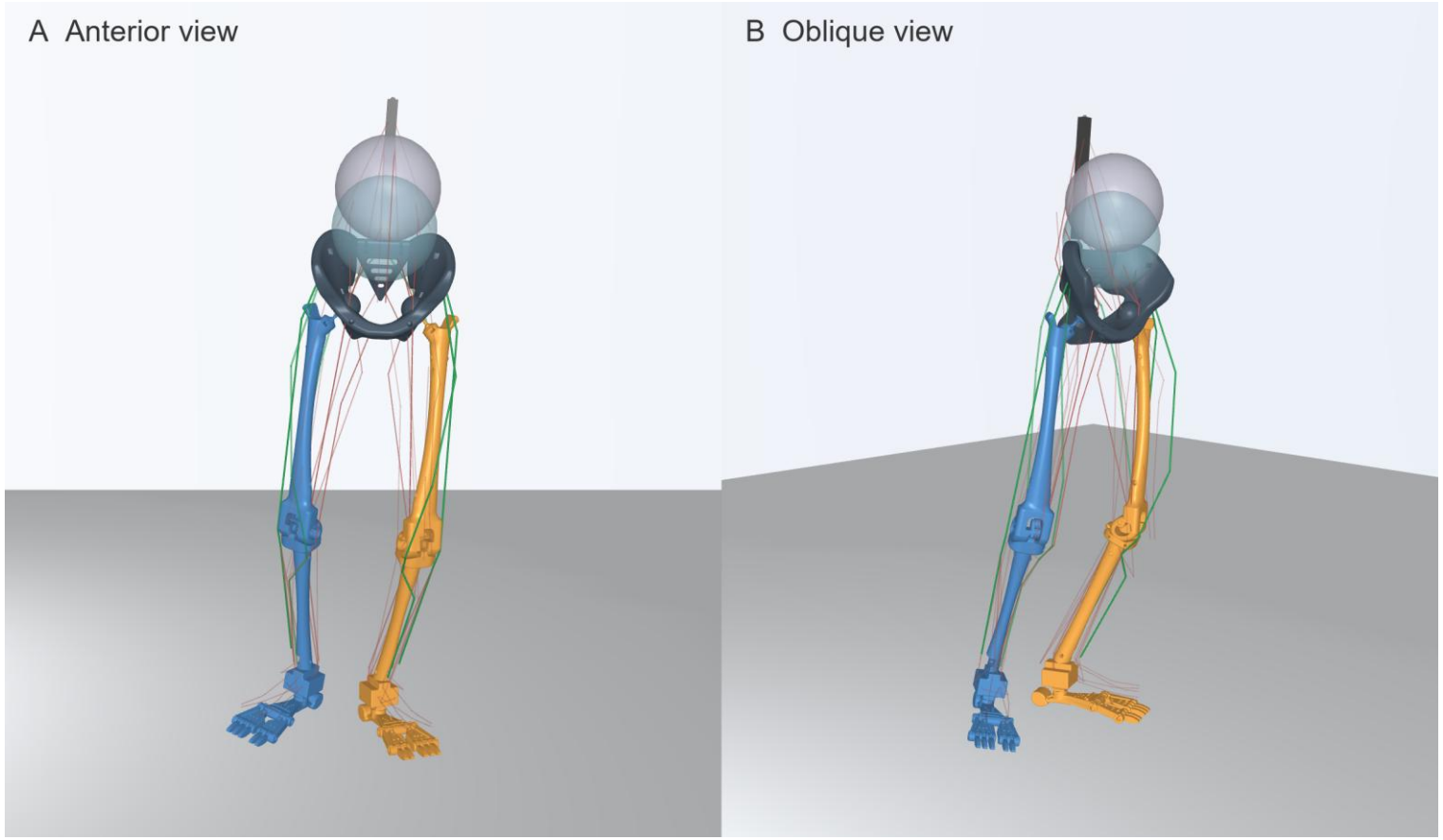
Hip timing differs

Knee timing differs

No fitted phase shift or amplitude scaling

10 cycles per leg · stance fractions 0.71 / 0.65 versus reference 0.61

Extra abductor control prevents crossing in the AH001 test



15–40 s	Off	On
Crossed states	67%	0%
Min. spacing	−38.3 mm	142.4 mm

54 reference-tension actuators · 262 neurons / 230 inputs

The evidence establishes specific capabilities within clear limits

Evidence level	What the result establishes	Remaining boundary
Measured hardware	Isometric torque capacity	Dynamic response and transfer
Neural demonstrations	Specified response and intervention effect	Circuit- and protocol-specific
Supported W2L / human models	Coupling and coordination tests	External support remains
AH001 integration	Hybrid allocation; PF/reflex currents cancelled	80% vertical + lateral/angular support

AH001: reference tensions, MN clamp, provisional feet, and ideal brackets

No harness-free walking has been demonstrated

Each contribution closes a different part of the original gap

Contribution	Decisive evidence
Resting-length-aware BPA model	Measured force across length, pressure, and contraction
Corrected knee model and routes	Lower held-out error; 24 of 24 physical targets met
Sensory evidence application	1,393 searchable, linked app records
Neuromechanical experiment tools	Matched circuits, feedback interventions, and IK overlays

The next experiments should test transfer and control authority

Open question	Next decisive check
Do torque corrections transfer?	Matched-pressure and dynamic-load tests
Do neural pathways control muscle output?	PF/reflex authority without MN current cancellation
Can the CAD robot sustain accepted stepping?	Verified mounts, alternating swing, and measured support

The robot’s design assumptions are now experimentally testable

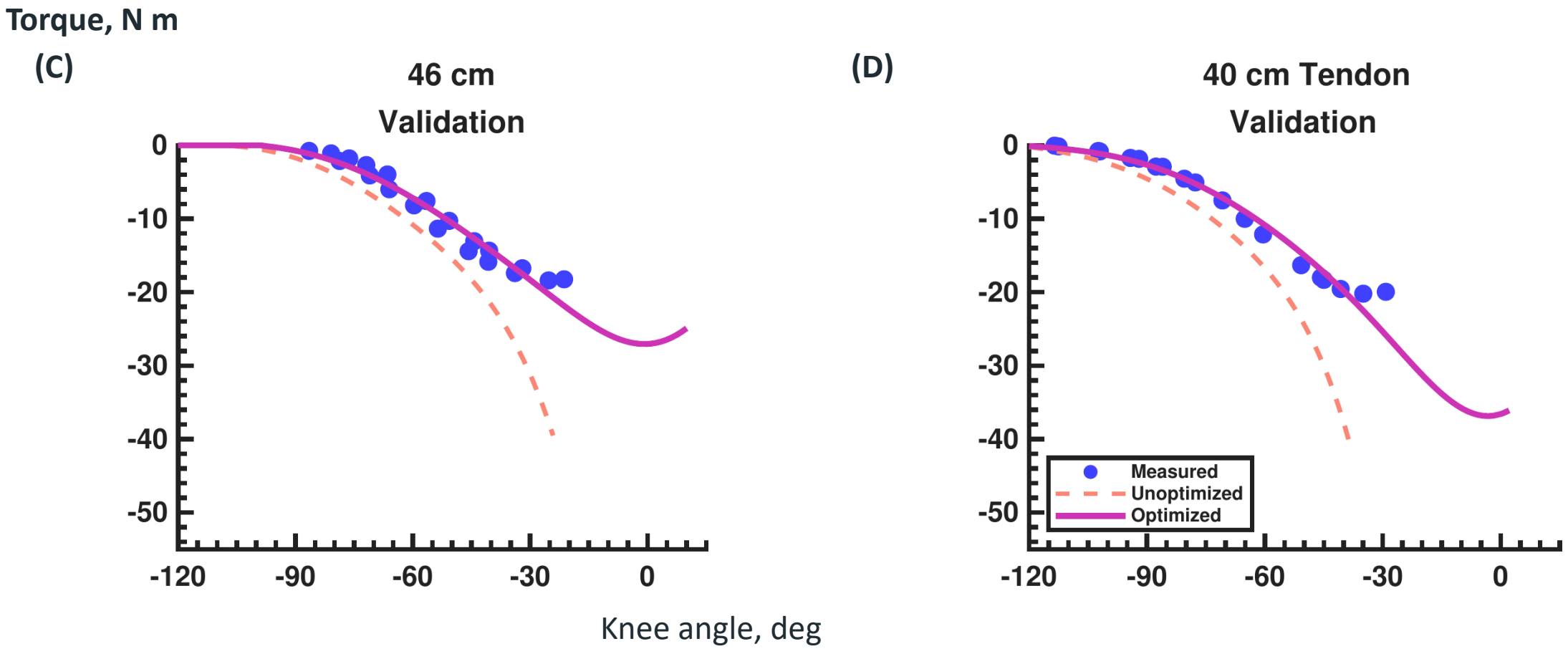
Force predictions grounded in measurement

24 of 24 physical knee targets met

Neural and mechanical hypotheses tested separately

Thank you · Questions

The held-out flexor traces show the corrected torque trend



Held-out mean RMSE: 7.588 to 1.490 N m

The adopted identification reserves tests in each configuration

Configuration	Training	Held out	Not used
Pinned flexor	2	2	1
Pinned extensor	3	3	3

Shared effective stiffness pair: $\chi_1 = 39.98$, $\chi_2 = 14.73$ kN/m

Physical dimensions differ from prediction settings

Actuator	Resting length, mm	Tendon length, mm
Flexor 1	561	28
Flexor 2	560	30
Extensor 1	828	92
Extensor 2	826	85

Prediction inputs: flexor 561.49 / 5.00 mm; extensor 827.74 / ≈92.49 mm

Stimulation duration changes the deletion interpretation

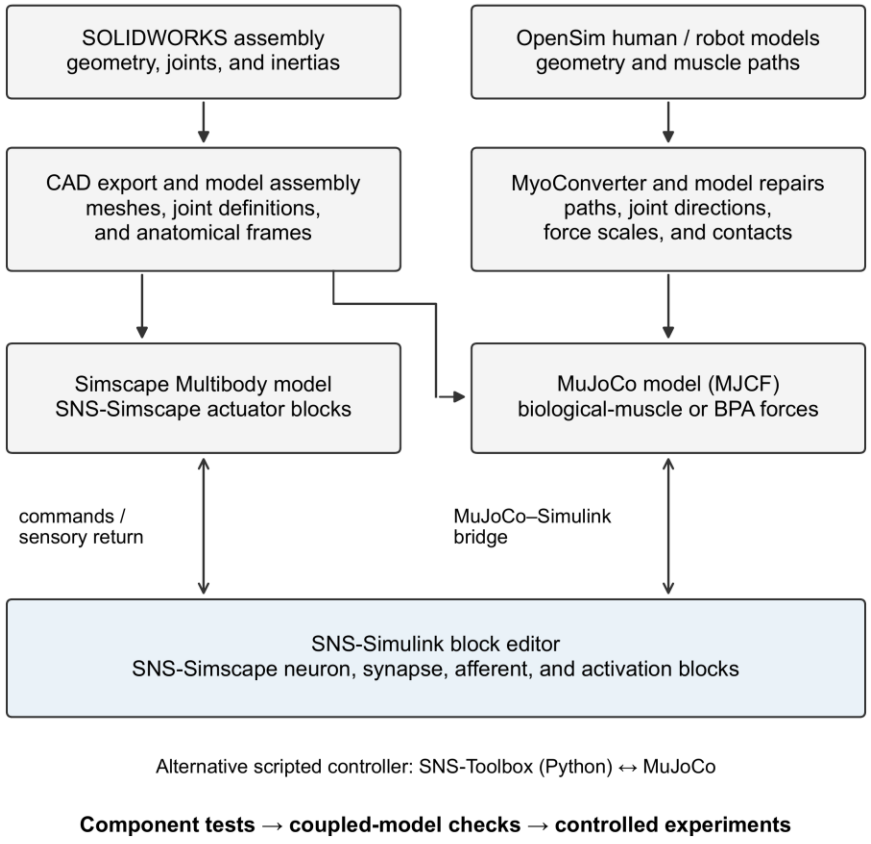
Protocol	RG shift, cycles	Motor shift, cycles
RG, two periods	0.211	0.211
PF, two periods	0.000	0.000
MN, two periods	0.000	0.000
PF, one second	0.000	0.208

stimulus_results_all.csv · Current Appendix D layer-stimulus protocols

AH001 separation control changes a matched diagnostic

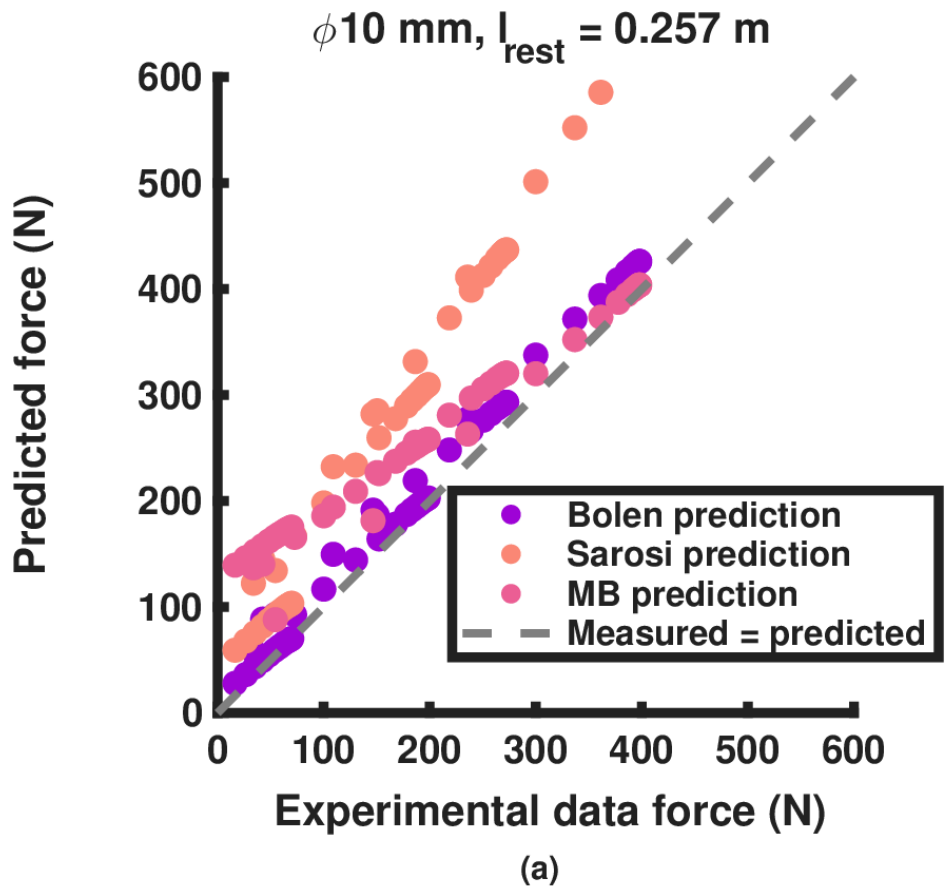
15–40 s measure	Control off	Control on
Reversed ankle ordering	67%	0%
Minimum separation	−38.3 mm	142.4 mm
Right TFL mean activation	22.3%	81.8%
Vertical assistance	80% BW	80% BW
Sampled opposite-leg penetration	2.700 mm	0 mm

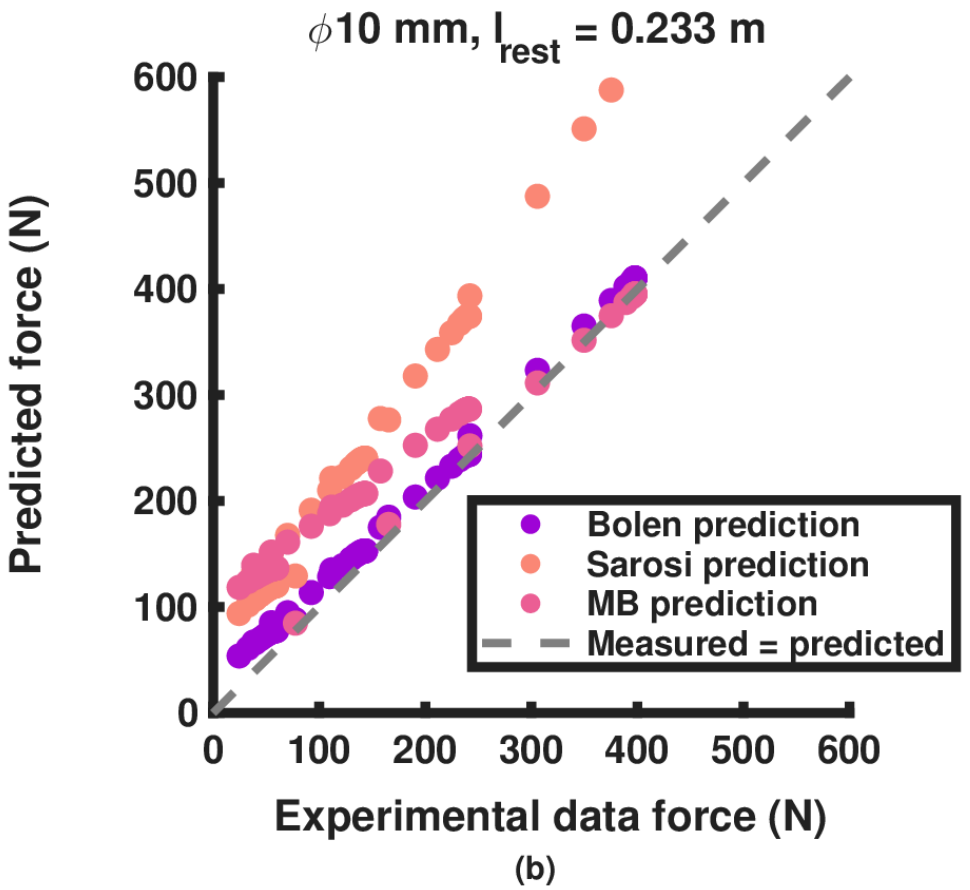
The model workflow supports distinct mechanical implementations



Command exchange, circuit agreement, and motion each require their own check

Reported force comparisons favor the model in the shown cases

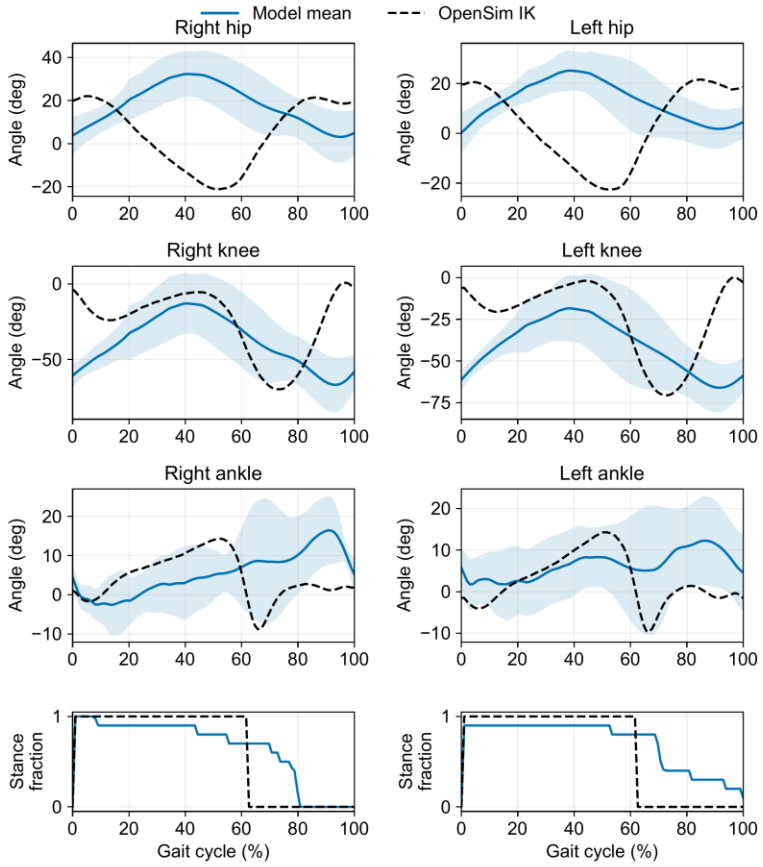




Measured-versus-predicted force at the two displayed actuator lengths

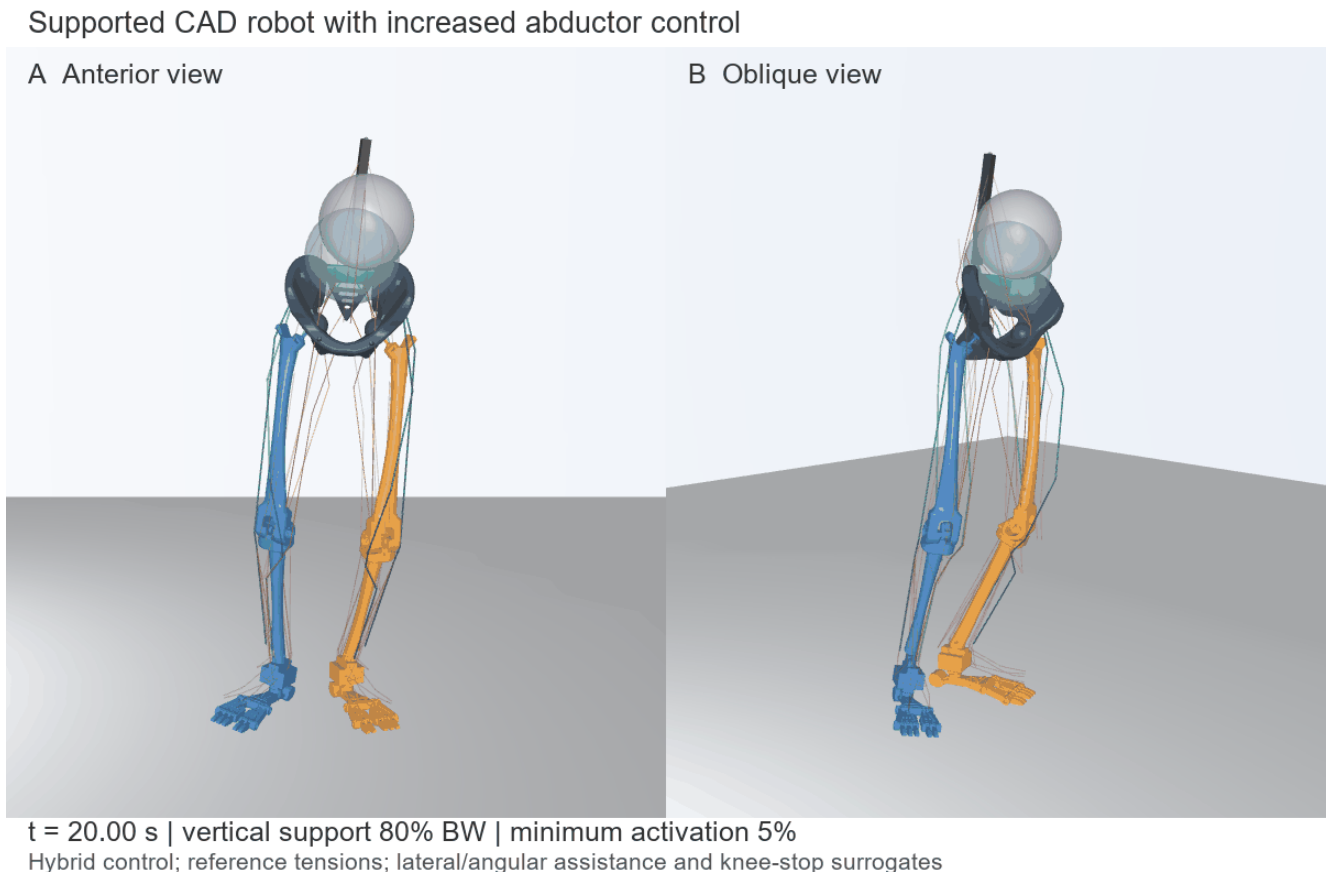
Bolen et al. (2026), Actuators 15(5):230 · Dissertation Ch. 3

The complete IK overlay also exposes contact differences



Model mean and standard deviation · OpenSim IK reference

The saved AH001 replay shows supported hybrid motion



Optional 12-second replay · same supported-control diagnostic

AH001_20261028/asset_provenance.json · saved 20–32 s interval